

An Analysis of Stochastic gradient descent

Stochastic gradient descent

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)$$

Stochastic gradient descent -- Part 1

$\bar{w} = \frac{1}{t} \sum_{i=0}^{t-1} w_i$. When optimization is done, this averaged parameter vector takes the place of w .

Stochastic gradient descent -- Part 2

A compromise between computing the true gradient and the gradient at a single sample is to compute the gradient against more than one training sample (called a "mini-batch") at each step. This can perform significantly better than "true" stochastic gradient descent described, because the code can make use of vectorization libraries rather than computing each step separately as was first shown in where it was called "the bunch-mode back-propagation algorithm". It may also result in smoother convergence, as the gradient computed at each step is averaged over more training samples. The convergence of stochastic gradient descent has been analyzed using the theories of convex minimization and of stochastic approximation. Briefly, when the learning rates η decrease with an appropriate rate, and subject to relatively mild assumptions, stochastic gradient descent converges almost surely to a global minimum when the objective function is convex or pseudoconvex, and otherwise converges almost surely to a local minimum. This is in fact a consequence of the Robbins–Siegmund theorem.

Stochastic gradient descent -- Part 3

where $x'w = x_1w_1 + x_2w_2 + \dots + x_pw_p$
 $x_{\{j\}}'w = x_{\{j1\}}w_{\{1\}} + x_{\{j,2\}}w_{\{2\}} + \dots + x_{\{j,p\}}w_{\{p\}}$
indicates the inner product. Note that x could have "1" as the first element to include an intercept. Classical stochastic gradient descent proceeds as follows:

Stochastic gradient descent -- Part 4

for $\Sigma(w) = \frac{1}{n} \sum_{i=1}^n (Q_i(w) - Q(w))(\sum_{i=1}^n Q_i(w) - Q(w))^T$
 $\left(\sum_{i=1}^n Q_i(w) - Q(w) \right)^2$
 $\left(\sum_{i=1}^n Q_i(w) - Q(w) \right)^T$ where dB_t denotes the Ito-integral with respect to a Brownian motion is a more precise approximation in the sense that there exists a constant $C > 0$ such that

Stochastic gradient descent -- Part 5

$$\max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - g(W_{k\eta})| \leq C\eta$$

Stochastic gradient descent -- Part 6

Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

Stochastic gradient descent -- Part 7

where i is uniformly sampled between 1 and n . Although theoretical convergence of this procedure happens under relatively mild assumptions, in practice the procedure can be quite unstable. In particular, when η is misspecified so that $I - \eta x_i x_i'$ has large absolute eigenvalues with high probability, the procedure may diverge numerically within a few iterations. In contrast, implicit stochastic gradient descent (shortened as ISGD) can be solved in closed-form as:

Stochastic gradient descent -- Part 8

subject to additional stochastic noise. This approximation is only valid on a finite time-horizon in the following sense: assume that all the coefficients Q_i are sufficiently smooth. Let $T > 0$ and $g: \mathbb{R}^d \rightarrow \mathbb{R}$ be a sufficiently smooth test function. Then, there exists a constant $C > 0$ such that for all $\eta > 0$

Stochastic gradient descent -- Part 9

$$w := w - \eta \nabla Q(w) = w - \frac{\eta}{n} \sum_{i=1}^n \nabla Q_i(w)$$

Stochastic gradient descent -- Part 10

Further reading Bottou, Léon (2004), "Stochastic Learning", Advanced Lectures on Machine Learning, LNAI, vol. 3176, Springer, pp. 146–168, ISBN 978-3-540-23122-6 Buduma, Nikhil; Locascio, Nicholas (2017), "Beyond Gradient Descent",

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Fundamentals of Deep Learning : Designing Next-Generation Machine Intelligence Algorithms, O'Reilly, ISBN 9781491925584 LeCun, Yann A.; Bottou, Léon; Orr, Genevieve B.; Müller, Klaus-Robert (2012), "Efficient BackProp", Neural Networks: Tricks of the Trade, Springer, pp. 9–48, ISBN 978-3-642-35288-1 Spall, James C. (2003), Introduction to Stochastic Search and Optimization, Wiley, ISBN 978-0-471-33052-3

Stochastic gradient descent -- Part 11

The scaling factor $\xi^* \in \mathbb{R}$ can be found through the bisection method since in most regular models, such as the aforementioned generalized linear models, function $q(\cdot)$ is decreasing, and thus the search bounds for ξ^* are $[\min(0, f(0)), \max(0, f(0))]$.

Stochastic gradient descent -- Part 12

The step size is denoted by η (sometimes called the learning rate in machine learning) and here $\eta := \{\}$ denotes the update of a variable in the algorithm. In many cases, the summand functions have a simple form that enables inexpensive evaluations of the sum-function and the sum gradient. For example, in statistics, one-parameter exponential families allow economical function-evaluations and gradient-evaluations. However, in other cases, evaluating the sum-gradient may require expensive evaluations of the gradients from all summand functions. When the training set is enormous and no simple formulas exist, evaluating the sums of gradients becomes very expensive, because evaluating the gradient requires evaluating all the summand functions' gradients. To economize on the computational cost at every iteration, stochastic gradient descent samples a subset of summand functions at every step. This is very effective in the case of large-scale machine learning problems.

Stochastic gradient descent -- Part 13

RMSProp (for Root Mean Square Propagation) is a method invented in 2012 by James Martens and Ilya Sutskever, at the time both PhD students in Geoffrey Hinton's group, in which the learning rate is, like in Adagrad, adapted for each of the parameters. The idea is to divide the learning rate for a weight by a running average of the magnitudes of recent gradients for that weight. Unusually, it was not published in an article but merely described in a Coursera lecture.